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Retrieval-Augmented Generation for LLM-based Recommender Systems: A Comprehensive Survey

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Abstract

Recommender systems are essential to real-world online services due to their substantial business value. Increasingly, large language models (LLMs) have been adopted for recommendation tasks, owing to their strong reasoning capabilities and their effectiveness in handling cold-start items. A large body of work has sought to adapt retrieval-augmented generation (RAG) for LLM-based recommender systems by retrieving information from external databases and integrating it into the LLM’s input to improve recommendation quality. Despite the rapid growth of the field, existing studies lack a clear taxonomy and a dedicated review of RAG methods for LLM-based recommender systems. To fill this gap, we present the first survey on this topic. First, we review the types of information used in LLM-based recommender systems, including target-user information and external databases for retrieval. Second, we characterize the design space of RAG for LLM-based recommender systems by outlining the key design choices at each stage of the pipeline: (1) retrieval, (2) augmentation, and (3) generation. Lastly, we discuss open questions for future research.

1 Introduction

Recommender systems play a crucial role in modern online platforms by helping users discover items of interest from an overwhelming number of available options (Zou and Sun, 2025; Peng et al., 2025). In practice, recommender systems enhance user satisfaction and drive business revenue.

Recently, significant efforts have been made to use large language models (LLMs) as recommender systems (Zhu et al., 2025a; Behdin et al., 2025). With strong reasoning and zero-/few-shot capabilities, LLM-based recommenders effectively

handle cold-start items and generate intuitive explanations that improve the interpretability of recommendation results (Zhang et al., 2025c).

To better adapt LLMs to recommendation tasks, retrieval-augmented generation (RAG) techniques have been increasingly employed (Wang et al., 2025b; Zhou et al., 2025). Typically, these approaches follow a three-stage pipeline: (1) retrieving recommendation-relevant knowledge from an external database, (2) augmenting the LLM’s input by incorporating the retrieved information, and (3) generating recommendations with the LLM based on the augmented input. In Figure 1, we present an example workflow of RAG for LLM-based recommender systems.

Despite growing interest in this area, existing surveys fall short of providing a systematic review and a comprehensive taxonomy regarding RAG for LLM-based recommendation. Closely related survey lines include (1) LLM-based recommender systems (Zhao et al., 2024b; Wang et al., 2024a) and (2) general RAG in natural language processing (NLP) (Fan et al., 2024; Zhao et al., 2024a). Existing surveys on LLM-based recommender systems rarely examine RAG in depth and therefore lack a systematic taxonomy of it, while existing RAG surveys in NLP focus on generic tasks and do not explicitly consider recommendation-specific factors, making their taxonomies less suitable for recommendation tasks.

To fill this gap, in this work, we present a timely survey that cover literature on RAG for LLM-based recommendation. To this end, we answer the five research questions provided below:

Q1. What information is available for users?

Q2. What information in the external database can be used for retrieval?

Q3. What retrieval strategies can be used?

Q4. What augmentation strategies can be used?

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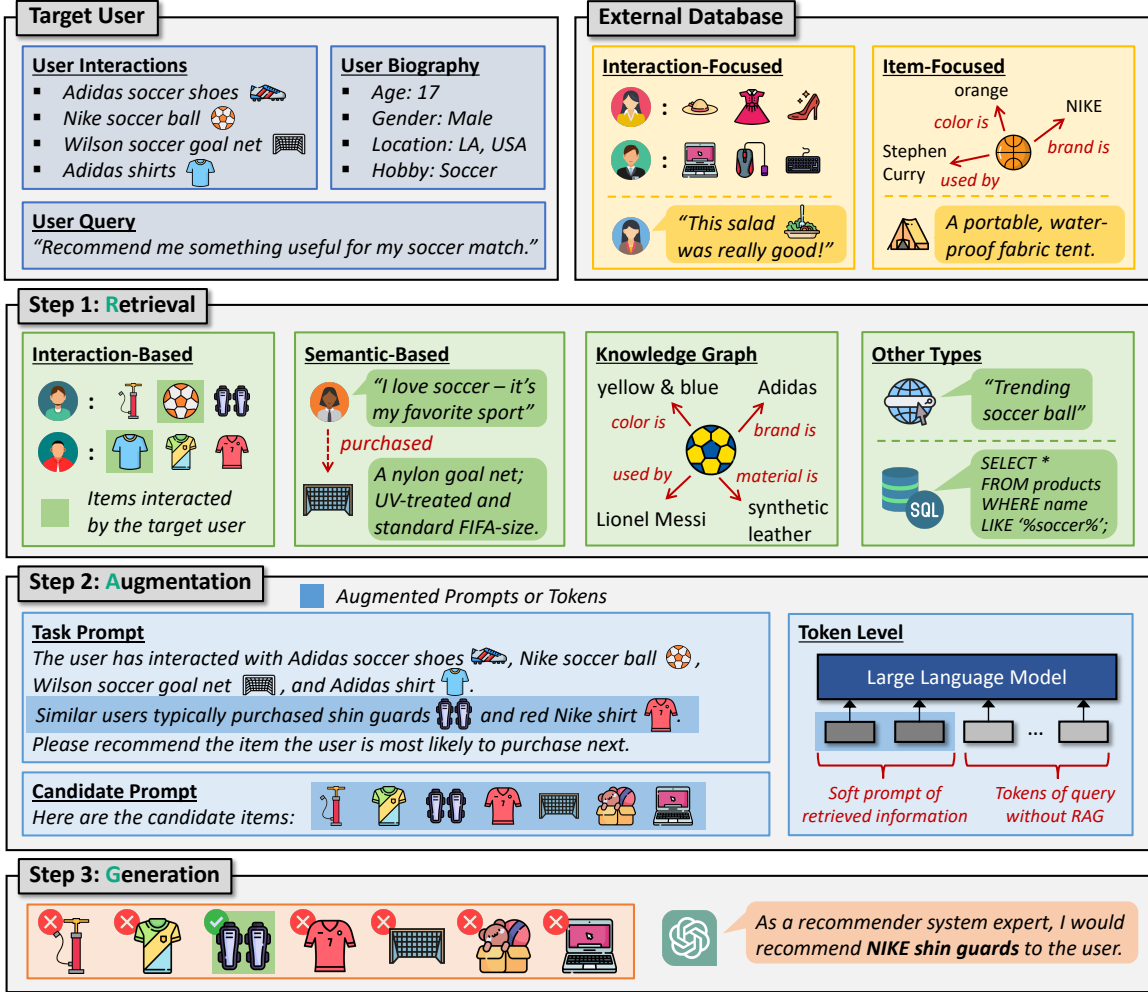


Figure 1: **Example workflow of retrieval-augmented generation (RAG) for LLM-based recommendation.** For a target user interested in soccer, the system may retrieve users with similar purchase behaviors, users with related query histories, and/or supplementary information about the items the target user has interacted with (Step 1). The retrieved information is then incorporated into the LLM’s input, either as natural-language context in the prompt or as soft tokens (Step 2). Finally, the LLM generates recommendations by ranking the candidate items (Step 3).

Q5. What generation strategies can be used?

For each question, we present a taxonomy of the design choices explored in prior work. A visual overview of these choices is provided in Figure 2.

The remainder of this survey is organized as follows. In Section 2, we review the information available for a target user and an external database (Q1 and Q2). In Section 3, we investigate retrieval strategies for obtaining recommendation-relevant knowledge from the database (Q3). In Section 4, we examine how to augment LLMs’ input with retrieved information (Q4), and in Section 5, we review how LLMs generate recommendation outputs (Q5). In Section 6, we discuss open questions in this field, and in Section 7, we conclude the survey.

2 User Information and Database

In this section, we review the types of information available for RAG in LLM-based recommender systems. Broadly, such information can be categorized into (1) information available for the target user and (2) information available in the external database for retrieval, which are discussed in Sections 2.1 and 2.2, respectively.

2.1 Target user information

The information available for a target user is essential for inferring the target user’s preferences, guiding both the (1) retriever in identifying relevant information and the (2) LLM-based recommender in generating recommendations. Such information can be categorized into three types: interaction-based, query-based, and biographical.

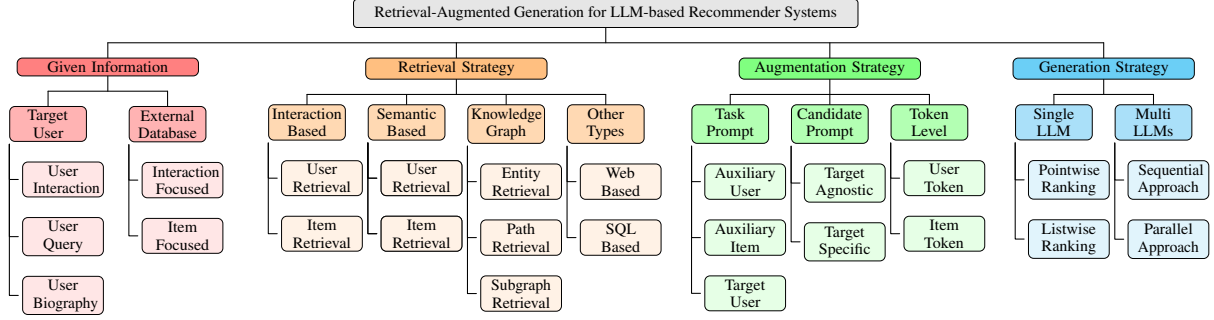


Figure 2: **Taxonomy of RAG for LLM-based recommender systems.** A summary of possible design choices for each process is provided.

Interaction-based. We can be provided with the items the target user has previously interacted with. Specifically, such information includes the items the target user has (1) clicked on (Xu et al., 2025a), (2) purchased (Wang et al., 2025b), (3) rated (Wu et al., 2024), and (4) reviewed in text (Wei et al., 2025), among others.

Query-based. We can also be provided with queries from the target user that reflect the user’s preferences. Specifically, the queries can take the form of (1) a single-turn request (e.g., “*recommend me a romance movie like Titanic*”) (Kemper et al., 2024), (2) a multi-turn conversation history between the target user and a conversational agent (Li et al., 2025a), or (3) multimodal queries that include images of items the target user is interested in (Cho et al., 2025).

Biographical. The target user’s biographical information, such as age, place of residence, or gender, can also be provided (Li et al., 2025b).

2.2 External database

The information contained in the external database serves as the primary source of retrievable knowledge, which helps LLMs provide higher-quality recommendations. This can be categorized into (1) interaction-focused and (2) item-focused.

Interaction-focused. We can be provided with user–item interaction data. Notably, the users considered here extend beyond the target user, encompassing interactions between other users and items in the database. Interactions include clicks, purchases, ratings, and textual reviews, among others, as discussed in Section 2.1. The key use cases include retrieving (1) users who exhibit interaction patterns similar to those of the target user (Wang and Lim, 2024), and (2) items purchased by users with similar behaviors to the target user (Kim et al., 2025), as elaborated in Section 3.1.

Item-focused. Additional item-related informa-

tion can be provided. Notably, this information goes beyond the item’s title and detailed description available without retrieval. A representative example is knowledge graph data, in which relational information about items is stored in a structured manner. The key use cases include (1) enriching an item’s description with information related to the corresponding item (Meng et al., 2025), and (2) explicitly matching item characteristics to the target user’s stated preferences (Yang et al., 2025) (e.g., retrieving romance movies for a user who requests a romance genre), as elaborated in Section 3.3.

3 Retrieval Strategies

In this section, we discuss retrieval strategies used in **RAG** for LLM-based recommendation. In the retrieval step, given a target user, we retrieve knowledge relevant to the user information (Section 2.1) from the external database (Section 2.2), which helps LLMs generate recommendations for the user. Such retrieval strategies can be categorized into four types: (1) interaction-based, (2) semantic-based, (3) graph-based, and (4) others.

3.1 Interaction-based retrieval

This category includes retrieval strategies that leverage user–item interactions stored in the external database. Overall, such methods aim to retrieve relevant users and/or items.

User retrieval. One line of methods retrieves users with interaction patterns similar to those of the target user. Retrieved users often serve as in-context examples that help LLMs capture the patterns of users with preferences similar to those of the target user. Among such approaches, one branch of work relies on heuristic rule-based strategies, such as measuring user similarity by counting the number of overlapping purchased items with the target user (Wang and Lim, 2024). Another branch of

this category employs collaborative filtering recommender systems (e.g., LightGCN (He et al., 2020)) to derive user representations from trained models and retrieve similar users based on representation-level similarity (Zhou et al., 2025).

Item retrieval. Another line of methods retrieves items that the target user is likely to interact with. Such items are commonly used (1) as candidate items that LLMs are prompted to rank for the target user and/or (2) as auxiliary information about items the target user has interacted with, providing additional context that improves recommendation quality. In this category, retrieval can be performed using heuristic rules, such as finding items that are frequently co-purchased with the target user’s previously purchased items (Kim et al., 2025), or using a collaborative-filtering-based recommender systems to retrieve the items they recommend for the target user (Zhu et al., 2025b).

3.2 Semantic-based retrieval

This category includes retrieval strategies that use semantic information to retrieve users and/or items semantically aligned with the target user.

User retrieval. One line of work focuses on retrieving users who are semantically similar to the target user, such as users whose prior conversations with the recommender chatbot are similar to the target user’s current conversation. Often, the retrieved information from similar users serves as in-context examples for the target user. A notable example is CRAVE (Zhu et al., 2025a), in which both the target user’s conversation and other users’ conversations in the database are encoded using a pretrained language encoder (e.g., Sentence-BERT (Reimers and Gurevych, 2019)), and the top- K most similar conversations are retrieved based on their embedding similarity.

Item retrieval. Another line of work focuses on retrieving items that are semantically aligned with the target user’s query, where the retrieved items often serve as candidates for LLM-based recommendation. To this end, both the user query and item descriptions are encoded using either sparse lexical models (Kieu et al., 2025) or pretrained language encoders (Kemper et al., 2024), and the top- K most similar items are retrieved according to their embedding similarity. In addition, retriever can use multimodal information, capturing not only textual semantics but also visual semantics from item images (Balachandran and Masum, 2025).

3.3 Knowledge-graph-based retrieval

This category includes retrieval strategies that leverage knowledge graphs, where relational data regarding the items are stored. Overall, such methods aim to retrieve relevant entities, paths, and subgraphs from the knowledge graphs.

Entity retrieval. First approach retrieves entities from the knowledge graph that are relevant to the target user, using them as candidate items for recommendation. A representative example is ColdRAG (Yang et al., 2025). Specifically, when the target user provides a query, ColdRAG selects entities that exhibit the greatest overlap with the query keywords. ColdRAG then starts from the identified entities and traverses the knowledge graph using a specialized LLM-based traversal strategy to retrieve item entities that can serve as candidates.

Path retrieval. Second approach retrieves paths from the knowledge graph as auxiliary information for items, helping the LLM better capture item-related knowledge that is beneficial for recommendation. A representative example is KGLA (Guo et al., 2025). Specifically, KGLA first locates the target user entity in the knowledge graph and links it to the item entities the target user has interacted with. KGLA then retrieves knowledge-graph paths that connect the target-user entity to candidate-item entities. Finally, the retrieved paths are incorporated into the LLM prompt as auxiliary information for recommendation.

Subgraph retrieval. A final approach retrieves subgraphs from the knowledge graph to provide the LLM with auxiliary information about item entities relevant to the target user, incorporating such items’ multi-hop relational information. A representative example is K-RagRec (Wang et al., 2025b). Specifically, K-RagRec first trains a graph neural network-based retriever that contrasts the user’s query representation with the k -hop subgraph representation of the ground-truth item within a knowledge graph. Then, at inference time, K-RagRec retrieves a fixed number of subgraph representations with high contrast scores with respect to the user query representation. The retrieved subgraph representations are incorporated into the LLM input token sequence, as elaborated in Section 4.3.

3.4 Others

Lastly, we discuss several other categories of retrieval strategies for LLM-based recommendation.

Web-based retrieval. This approach retrieves in-

formation relevant to the target user from the web. A notable example is WebRec (Zhao et al., 2025). Specifically, given the target user’s item interaction history, WebRec first prompts an LLM to extract keywords that reflect the user’s preferences. Then, WebRec searches for related keywords on the web, retrieves relevant webpages associated with those keywords (e.g., social media posts). This information is included in the input prompt to provide auxiliary information about the target user.

SQL-based retrieval. This approach retrieves information relevant to the target user from a tabular database using SQL queries. A notable example is TalkPlay-Tools (Doh et al., 2025). For example, for a user query requesting fast music, TalkPlay-Tools uses SQL to query a tabular database of track metadata and retrieves songs whose BPM exceeds a specified threshold.

4 Augmentation Strategies

In this section, we discuss the augmentation strategies used in RAG for LLM-based recommendation. In the augmentation step, the retrieved information is incorporated into LLM inputs. Typically, augmentation follows one of three strategies: (1) task-prompt augmentation, which incorporates auxiliary context to help LLMs make more accurate recommendations; (2) candidate-prompt augmentation, which includes retrieved candidate items and/or adds auxiliary information about each candidate; (3) token-level augmentation, which injects vector representations of retrieved information into input token sequences as soft prompts.

4.1 Task-prompt augmentation

This category includes prompt augmentation methods that incorporate auxiliary information relevant to the target user to help LLMs generate more accurate recommendations. Broadly, such approaches can be categorized into three types: (1) incorporating auxiliary user information, (2) incorporating auxiliary item information, and (3) augmenting the target user’s information.

Auxiliary user addition. This approach incorporates information about users similar to the target user into LLMs’ recommendation prompt as reference context for generating recommendations. Note that similar users can be retrieved using the strategies described in Sections 3.1 and 3.2. Retrieved user information can be incorporated into recommendation prompts using different prompt-

engineering strategies. For example, when the purchase histories of similar users are retrieved, one can directly include the full history in a prompt (e.g., “A user similar to the target user has the following purchase history: (retrieved history).”) (Zhang et al., 2025b). Alternatively, the retrieved histories can be incorporated in an in-context example format, where the last item is held out and presented as the ground-truth item corresponding to the preceding interaction history (e.g., “Given the following interaction history (retrieved history without the last item), the correct next item is (last item of the retrieved history).”) (Xu et al., 2025c). Example prompts of respective cases are in Appendix A.3.

Auxiliary item addition. Unlike approaches that retrieve and incorporate similar user information, this approach incorporates item-level information relevant to the target user. A representative example is ItemRAG (Kim et al., 2025). Specifically, ItemRAG retrieves co-purchased items for each item in the target user’s purchase history. Then, ItemRAG refines each item’s description with the retrieved co-purchased items (e.g., “This item is co-purchased with (descriptions of retrieved items).”), thereby injecting collaborative information that is often beneficial for recommendation.

Target user augmentation. The last branch augments the target user’s interaction history to reduce noise within it. A representative example is ReACT (Hu et al., 2026). ReACT computes similarity scores for each item pair within the target user’s interaction history by measuring how often that pair co-appears in other users’ histories. ReACT then augments the history by excluding items whose average similarity to the other items in the history falls below a predefined threshold.

4.2 Candidate-prompt augmentation

Note that candidate items refer to the set of items provided to an LLM, from which it ranks and selects items to recommend to the target user. Alongside the original metadata associated with each candidate item, many approaches integrate additional retrieved information to enhance their descriptions. Typically, these augmentation methods can be divided into (1) target-user-agnostic augmentation and (2) target-user-specific augmentation.

Target-agnostic augmentation. This approach augments candidate item descriptions with target-user-agnostic information, applying the same augmentation to a given item across all users. Since

such information can be precomputed and coupled with items offline and cached, this approach enables fast retrieval during inference. For example, the retrieved information may include relational facts about each candidate item from a knowledge graph (Meng et al., 2025) or summaries of users who have interacted with the corresponding item (Wu et al., 2024).*

Target-specific augmentation. This approach augments candidate item descriptions with target-user-specific information, such that the retrieved content associated with the same item differs across users. Methods that provide the ranking scores from non-LLM-based recommender systems—typically collaborative filtering models—for candidate items with respect to the target user fall into this category (Luo et al., 2025a; Oak et al., 2025). Such scores help LLMs leverage knowledge from external recommender systems.

4.3 Token-level augmentation

Note that the above augmentation branches (Sections 4.1 and 4.2) incorporate retrieved information into LLMs’ input prompts in natural-language form. In contrast, specialized vector representations of the retrieved information can be injected into LLMs’ input sequences as soft prompt tokens. Although this approach requires an additional training stage to align the retrieved representations with LLMs’ token space, it typically results in shorter input sequences than prompt-based augmentation, enabling more efficient inference. Typically, such soft prompts can be categorized into (1) user-related tokens or (2) item-related tokens.

User-related token. These approaches incorporate user-related soft prompts, often derived from target-user representations learned by non-LLM-based recommender systems. Such representations compactly encode the collaborative information about the target user, typically as a small number of tokens (sometimes even a single token). The representative example is CoLLM (Zhang et al., 2025d). CoLLM trains a collaborative filtering model (e.g., SASRec (Kang and McAuley, 2018)) to obtain a target-user embedding. It then projects the embedding via an MLP into an LLM’s token embedding space and concatenates the resulting soft prompt to-

kens with the input token sequence from the natural-language recommendation prompt.

Item-related token. These approaches incorporate item-related soft prompts, often derived from items relevant to the target user (e.g., items the target user is likely to purchase). Such prompts compactly encode auxiliary information about the corresponding item as a vector representation. The representative example is K-RagRec, whose retrieval strategy is presented in Section 3.3. After retrieving the subgraph embeddings of item entities relevant to the target user, K-RagRec projects these embeddings into an LLM’s token embedding space using a learnable projection head. Subsequently, the projected embeddings are added as soft prompt tokens to the LLM’s input sequence. Note that, in addition to embeddings from knowledge graphs, embeddings of user-purchased items or candidate items from collaborative filtering models are also widely adopted (Liu et al., 2025b; Wang et al., 2025a).

5 Generation Strategies

In this section, we examine generation strategies for LLM-based recommender systems. At this stage, an LLM produces recommendation results for the target user based on augmented inputs. We categorize methods into (1) single-LLM-based approaches and (2) multi-LLM-based approaches.

5.1 Single-LLM-based approaches

This category comprises approaches that use a single recommender LLM for the entire pipeline. Such approaches can be categorized into two types based on how they formulate the recommendation task: pointwise ranking and listwise ranking.

Pointwise ranking. This branch consists of methods that prompt an LLM to estimate the likelihood that a target user will interact with a given single candidate item, and then recommend the top- K items with the highest estimated likelihood. This approach allows incorporating substantial information about a single candidate item, as each prompt is dedicated to one candidate. It thus typically fits within the LLM context-length limit, even when the retrieved information is extensive (Lee et al., 2024). A representative example is OptiRAG-Rec (Zhou et al., 2025). Given a recommendation prompt that includes retrieved information and asks whether to recommend the candidate item, OptiRAG-Rec uses the probability that the LLM outputs the token ‘Yes’ as the recommendation score.

*In a broad sense, approaches that first retrieve candidate items from the full item pool using a non-LLM recommender system and then prompt the LLM to rank the retrieved candidates can be regarded as RAG. Within this group, we classify methods that represent each candidate item solely by its own metadata as target-user-agnostic augmentation in Table 1.

Reference	Method	Target user (§ 2.1)			Database (§ 2.2)		Retrieval (§ 3)				Augmentation (§ 4)			Generation (§ 5)	
		Inter.	Query	Bio.	Inter.	Item	Inter.	Semantic	KG	Others	Task	Cand.	Token	Single	Multi
Yang et al. (2023)	PALR	✓			✓		✓					✓			✓
Wang et al. (2024b)	RecMind	✓	✓		✓	✓	✓	✓			✓	✓			✓
Yang and Chen (2024)	ReFICR	✓			✓	✓	✓				✓	✓			✓
Shen et al. (2024)	URLLM	✓			✓	✓	✓	✓			✓	✓			✓
Zeng et al. (2024)	GPT-FedRec	✓			✓	✓	✓	✓			✓			✓	
Wang et al. (2024c)	MACRec	✓		✓	✓	✓	✓	✓			✓	✓			✓
Rao and Lin (2024)	RAMO		✓			✓		✓				✓		✓	
Lin et al. (2024)	ReLLa	✓		✓		✓		✓			✓				✓
Maes et al. (2024)	GenUI (ne)	✓				✓		✓			✓				✓
Wu et al. (2024)	CoRAL	✓			✓	✓	✓			✓	✓	✓			✓
Tan et al. (2024)	Per-Pcs	✓			✓		✓						✓	✓	
Xie et al. (2024)	NBCRS		✓			✓		✓			✓			✓	
Kemper et al. (2024)	RA-Rec		✓			✓		✓			✓			✓	
EL Maazouzi et al. (2024)	EL’s RAG [†]		✓			✓		✓				✓		✓	
Wang and Lim (2024)	Wang’s RAG [†]	✓			✓	✓	✓	✓			✓			✓	
Dehbozorgi et al. (2024)	Dehbozorgi’s RAG [†]		✓			✓		✓				✓		✓	
Cho et al. (2025)	MARC		✓			✓		✓	✓			✓			✓
Tourani et al. (2025)	LLMGenQC	✓			✓		✓	✓				✓			✓
Balachandran and Masum (2025)	VisioRAG		✓			✓		✓				✓			✓
Meng et al. (2025)	KERAG_R	✓			✓	✓	✓			✓	✓	✓		✓	
Zhu et al. (2025b)	CRAG		✓		✓		✓	✓			✓	✓			✓
Yang and Wang (2025)	CTS	✓				✓		✓				✓		✓	
Wang et al. (2025b)	K-RagRec	✓				✓			✓				✓	✓	
Maragheh et al. (2025)	ARAG	✓				✓		✓			✓	✓			✓
Wei et al. (2025)	PrefRAG	✓			✓	✓	✓				✓	✓		✓	
Alamri et al. (2025)	RankRAG		✓			✓		✓				✓		✓	
Tandon and Banerjee (2025)	RAG-C-TRS		✓			✓		✓				✓			✓
Xu et al. (2025a)	RALLRec	✓		✓	✓	✓	✓	✓			✓				✓
Zhang et al. (2025a)	SarRec	✓			✓	✓	✓	✓				✓		✓	
Shehmir and Kashef (2025)	PromptRec			✓	✓	✓	✓	✓			✓			✓	
Zheng et al. (2025)	RAGPR		✓			✓		✓			✓			✓	
Kieu et al. (2025)	KALM4Rec		✓		✓	✓		✓			✓			✓	
Yang et al. (2025)	ColdRAG	✓				✓				✓	✓	✓			✓
Zhou et al. (2025)	OptiRAG-Rec	✓	✓		✓		✓				✓			✓	
Ma et al. (2025)	AgenticRAG	✓	✓			✓		✓			✓				✓
Zhu et al. (2025a)	CRAVE		✓		✓		✓				✓				✓
Li et al. (2025a)	ChatCRS		✓		✓	✓			✓		✓				✓
Zhao et al. (2025)	WebRec	✓			✓	✓	✓			✓	✓	✓			✓
Luo et al. (2025a)	RecRanker	✓			✓	✓	✓				✓	✓		✓	
Zhang et al. (2025b)	AdaptRec	✓			✓	✓	✓				✓				✓
Azizi and Koochaki (2025)	LLAMARec-KG-RAG	✓	✓		✓	✓	✓		✓		✓	✓		✓	
Liu et al. (2025a)	CoT-Rec	✓			✓	✓	✓					✓			✓
Petrucelli et al. (2025)	Petrucelli’s RAG [†]	✓			✓	✓	✓		✓		✓			✓	
Carraro et al. (2025)	Carraro’s RAG [†]	✓	✓		✓	✓	✓	✓		✓	✓				✓
Kim et al. (2025)	ItemRAG	✓			✓	✓	✓	✓			✓	✓			✓
Luo et al. (2025b)	RALLRec+	✓			✓	✓	✓	✓			✓	✓			✓
Askar and Baizal (2025)	Askar’s RAG [†]		✓			✓		✓			✓			✓	
Qiu et al. (2025)	G-CRS		✓			✓		✓	✓		✓				✓
Liu et al. (2025b)	Cora	✓			✓		✓						✓	✓	
Wang et al. (2025a)	FACE	✓	✓		✓	✓	✓						✓		✓
Hu et al. (2026)	ReACT	✓	✓		✓	✓	✓				✓			✓	
Hui et al. (2026)	MATCHA		✓			✓		✓				✓			✓

Table 1: **Summary of RAG methods for LLM-based recommender systems.** Inter., Bio., KG, and Cand. denote interaction, biography, knowledge graph, and candidate, respectively. Each checkmark (✓) indicates that the RAG method in the corresponding row employs the technique specified in the corresponding column. [†] indicates that the corresponding method is unnamed in the original paper; we refer to it as “(first author)’s RAG”.

Listwise ranking. This branch consists of methods that prompt an LLM to directly rank candidate items in a single pass. Therefore, this approach requires fewer LLM inference calls per target user and enables the LLM to consider multiple candidates jointly within the same prompt. A representative example is KALM4Rec (Kieu et al., 2025). Given a set of retrieved candidate items, KALM4Rec prompts an LLM to rank candidate

items according to their likelihood of being purchased by the target user and recommends the top- K ranked items.

5.2 Multi-LLM-based approaches

This category comprises methods that use multiple LLMs with distinct roles throughout the pipeline. Such generation methods can be categorized into two types: (1) parallel-arrangement-based methods

and (2) sequential-arrangement-based methods.

Parallel arrangement. This branch includes methods in which multiple LLMs are arranged in parallel, each serving a distinct role and producing outputs that are forwarded to a final LLM recommender. Since the LLMs operate in parallel, the processes can be executed concurrently, thereby reducing the overall time cost. A representative example is ARAG (Maragheh et al., 2025), which employs two distinct LLMs: the first analyzes the target user’s interests, while the second evaluates how well the user profile aligns with each candidate item. Lastly, a final LLM generates recommendations based on the outputs of the first two LLMs.

Sequential arrangement. This branch includes methods in which multiple LLMs are arranged sequentially, each performing a designated role. Often, this structure enables one LLM to verify the output of another, thereby producing more accurate recommendation results (Hui et al., 2026; Li et al., 2025a). A representative example is CRAVE (Zhu et al., 2025a). Given a user query, CRAVE first generates recommendations with a CoT agent. The CoT agent’s output is then passed to a debater agent, which proposes alternative recommendations that differ from the CoT output by reasoning from a different perspective. Finally, a critic agent takes both the CoT and debater outputs as input and selects which recommendation(s) to present to the user. Importantly, each agent is augmented with role-specific collaborative experiences retrieved from other users.

6 Open Questions

In this section, we explore open questions in RAG for LLM-based recommender systems.

6.1 Efficiency improvement

Typically, augmenting a prompt with retrieved information increases its length, thereby introducing additional computational overhead to LLMs. Given that low latency is a critical requirement of recommender systems in practical deployments, it is essential to reduce such computational overhead. In the general Q&A domain, various adaptive RAG techniques have been proposed to improve efficiency, such as performing retrieval and augmentation only when necessary (Jeong et al., 2024; Tang et al., 2025). However, because such difficulty classification is tailored to NLP tasks, these techniques cannot be straightforwardly extended

to recommendation settings. Therefore, to enable practical deployment of RAG for LLM-based recommendation in real-world settings, it is important to develop specialized adaptive RAG techniques.

6.2 Privacy preservation

As described in Section 3, methods that retrieve users relevant to the target user often include information about those retrieved users, such as their click and purchase histories. However, such information often contains personal data about users. During generation, the LLM may unintentionally incorporate such information into its responses for the target user, raising privacy concerns due to the risk of personal information leakage. Therefore, addressing these privacy concerns is essential for the practical deployment of user-retrieval-based approaches. One promising direction is to replace real users in the retrieval pool with synthetic users (Wang and Lim, 2024).

6.3 Diversity improvement

When only highly similar items are recommended to a target user, the user may experience boredom, making such recommendations undesirable in practical recommender system applications. Despite this importance, most RAG methods for LLM-based recommender systems primarily consider how well retrieved items match the target user, but do not consider the diversity of the retrieved information. Moreover, existing diversity-aware RAG pipelines primarily focus on semantic information (Rezaei and Dieng, 2025), without leveraging information specific to recommender systems, such as user–item interactions. Therefore, investigating diversity-aware RAG methods tailored to LLM-based recommender systems is a promising direction for real-world deployment.

7 Conclusion

In this work, we present a comprehensive survey of RAG techniques for LLM-based recommender systems. We first review the available information about the target user and the external database. We then develop a taxonomy of key design choices across retrieval augmentation and generation strategies. Finally, we discuss open problems and promising directions for future research.

Our key takeaways include (1) the importance of RAG for integrating LLM capabilities into recommendation; (2) the need to specialize RAG to task-specific properties and data beyond general NLP

tasks; and (3) remaining challenges for practical systems, including efficiency, privacy preservation, and recommendation diversity.

Limitations

In this section, we explore several potential limitations of our survey.

Lack of benchmarking experiments. While this survey provides a comprehensive taxonomy and literature review of RAG methods for LLM-based recommender systems, it does not include a unified benchmark across diverse RAG approaches. This is mainly because (1) evaluation settings vary substantially across methods, spanning tasks such as conversational recommendation and sequential recommendation, and (2) many methods do not provide reproducible code. Therefore, developing a comprehensive and user-friendly benchmark for RAG in LLM-based recommender systems would be a valuable direction for future work.

Lack of backbone LLM effect analysis. Xu et al. (2025b) demonstrate that the performance of LLM-based recommenders significantly varies depending on the backbone LLM used. Therefore, the effectiveness of RAG methods is expected to vary depending on the backbone LLM. However, since existing studies have not provided a systematic analysis of this issue, we were unable to offer a comprehensive discussion in this survey. Accordingly, a systematic investigation of the interaction between RAG techniques and backbone LLMs would be a promising direction for future research.

Ethical Considerations

Since this survey covers only publicly available literature, we believe our work does not raise ethical concerns related to licensing or copyright infringement. Moreover, since our work does not involve data collection, dataset construction, model deployment, or human-subject experimentation, it does not present the ethical risks typically associated with these activities.

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A Prompt Examples

In this section, we provide example prompt formats for RAG in LLM-based recommendation.

A.1 Naive LLM-based recommendation prompt

Naive LLM-based Recommendation

Role: You are a recommender system.

Target user purchase history:
[User’s interaction history]

Candidate items:
[Candidate items]

Task: By considering all the inputs, rank the candidate items from most likely to least likely to be purchased by the user.

Figure 3: Example of naive LLM-based recommendation prompt.

A.2 Auxiliary user addition prompt

Auxiliary User Addition

Instruction: Based on the product of the input user has reviewed, deduce if the user will like the mentioned product.

Example:

[Similar user’s reviews]

Input:

[Target user’s interaction history]

[Target item]

You should ONLY tell me yes or no.

Figure 4: Example of auxiliary user addition prompt (Zhou et al., 2025).

A.3 Auxiliary item addition prompt

Auxiliary Item Addition

Role: You are a recommender system.

Target user purchase history:

[User’s interaction history]

Candidate items:

[Candidate items]

Additional item information:

[Additional item description (e.g., co-purchase summary)]

Task: By considering all the inputs, rank the candidate items from most likely to least likely to be purchased by the user.

Figure 5: Example of auxiliary item addition prompt (Kim et al., 2025).

B Benchmark Datasets

In this section, we explore benchmark datasets that are widely used in the research of LLM-based recommender systems.

B.1 Amazon review dataset

This dataset contains reviews and ratings that users have assigned to items on the Amazon website (Hou et al., 2024). The dataset encompasses reviews across diverse domains, including Sports & Outdoors, Toys & Games, Beauty & Personal care, Arts, Crafts & Sewing, CDs & Vinyls, among others. The dataset is commonly used as a source of user–item interaction data.

B.2 MovieLens dataset

This dataset contains ratings that users have assigned to movies on the MovieLens website (Harper and Konstan, 2015). The dataset includes movie ratings collected across a wide range of users and items, and some versions also provide additional metadata such as movie genres and user-generated tags. The dataset is commonly used as a source of user–item interaction data.

B.3 Yelp dataset

This dataset contains reviews and ratings that users have assigned to local businesses on the Yelp website (Yelp, 2023). The dataset includes reviews across diverse business categories, including restaurants and shopping, among others. Some versions also provide additional metadata such as business attributes, tips, check-ins, and user social network information. The dataset is commonly used as a source of user–item interaction data.

B.4 LastFM dataset

This dataset was obtained from the Last.fm online music platform (LastFM, 2011). It contains users’ listening records on music artists, along with tag assignments and social relations derived from user friendship links. The dataset is commonly used as a source of user–item interaction data.

B.5 Steam dataset

This dataset contains user interactions with games collected from the Steam platform (Kang and McAuley, 2018). It contains users’ implicit feedback signals, such as playtime and review activity, along with game metadata, including title, genre, developer, price, and release information. The dataset is commonly used as a source of user–item interaction data.

B.6 GoodReads datasets

This dataset contains user interactions with books collected from the Goodreads platform (Wan and

McAuley, 2018). It includes multiple types of user feedback, such as shelving, reading, rating, and recommendation signals, and some versions additionally provide rich book metadata, including authors, series, editions, publishers, page counts, languages, similar books, and user-generated shelf names. The dataset is commonly used as a source of user–item interaction data.

the taxonomy we developed.

B.7 BookCrossing dataset

This dataset contains book ratings submitted by users on the Book-Crossing platform (Ziegler et al., 2005). The dataset includes explicit ratings on a scale of 1 to 10, and some versions also provide additional metadata such as book title, author, publisher, and year of publication. The dataset is commonly used as a source of user–item interaction data in book recommendation.

B.8 Reddit dataset

This dataset contains recommendation-related conversations collected from users on the Reddit platform (He et al., 2023) under movie-seeking topics. It consists of naturally occurring dialogs from movie-related subreddits, where users seek and offer movie recommendations in real-world discussion settings. The dataset is commonly used as a source for conversational recommender systems.

B.9 Redial dataset

This dataset contains dialogues in which users recommend movies to each other (Li et al., 2018). It consists of multi-turn two-party conversations between a recommendation seeker and a recommender, where movie suggestions are refined based on the seeker’s preference and feedback. The dataset is commonly used in research on conversational recommender systems.

B.10 Inspired dataset

This dataset contains human-human recommendation dialogues collected in a setting for movie recommendation (Hayati et al., 2020). It consists of multi-turn conversations between a seeker and a recommender, where the recommender elicits preferences and suggests movies during the dialogue. The dataset is commonly used in research on conversational recommender systems.

C AI Assistant Usage

We used large language models only to refine the writing of this paper and to provide feedback on